

GEOGRAPHY 32 - INDUSTRIES AND INDUSTRIAL REGIONS - LEARNER-FIRST V2

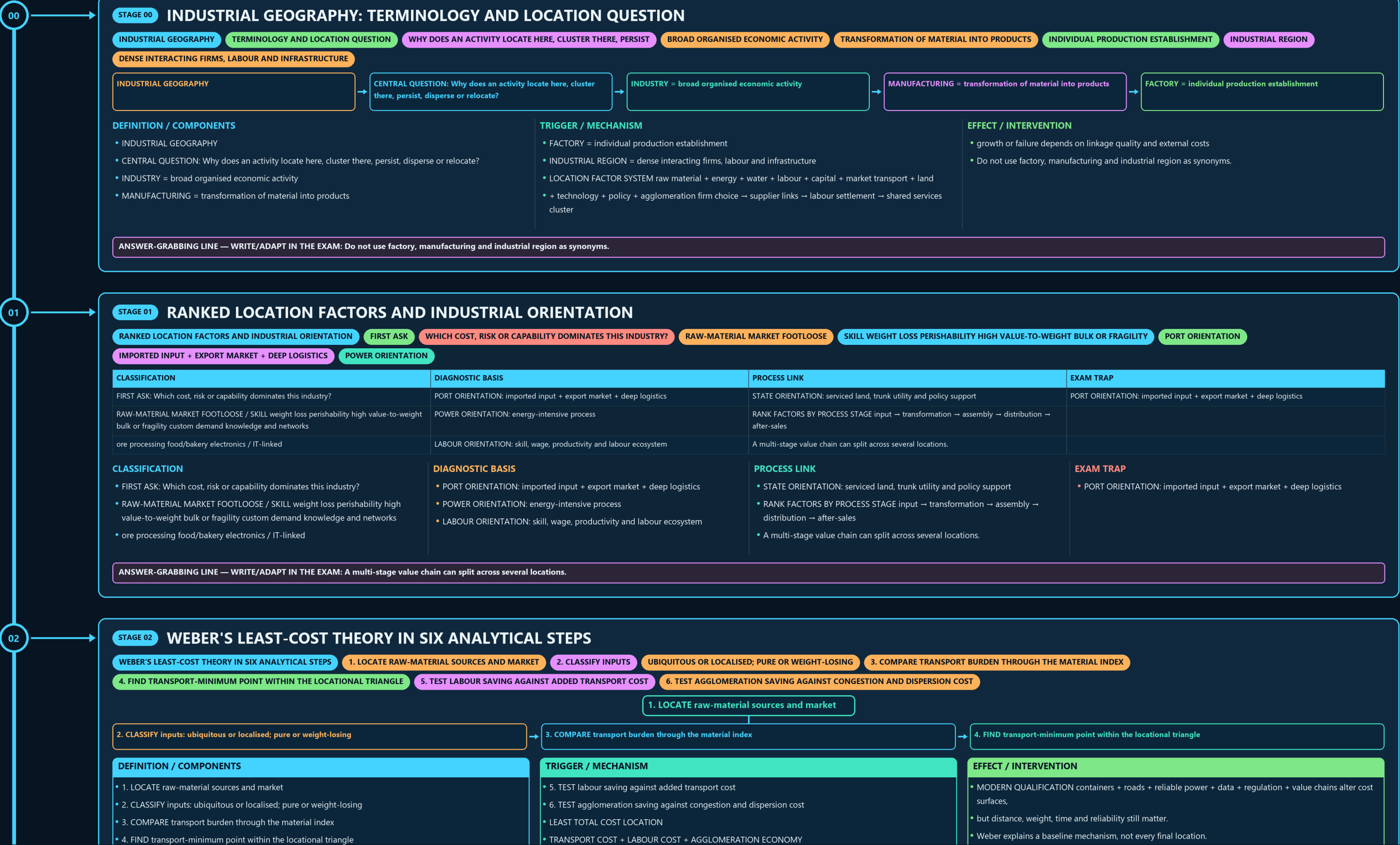
INDUSTRIAL GEOGRAPHY: TERMINOLOGY AND → RANKED LOCATION FACTORS AND → WEBER'S LEAST-COST THEORY IN → MAJOR WORLD INDUSTRIAL REGIONS → INDIA'S LAYERED INDUSTRIAL MAP → AGGLOMERATION, DEGLOMERATION, INERTIA AND

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g11 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



- RAW-MATERIAL MARKET FOOTLOOSE / SKILL weight loss perishability high value-to-weight bulk or fragility custom demand knowledge and networks
- ore processing food/bakery electronics / IT-linked

- POWER ORIENTATION: energy-intensive process
- LABOUR ORIENTATION: skill, wage, productivity and labour ecosystem

- RANK FACTORS BY PROCESS STAGE input → transformation → assembly → distribution → after-sales
- A multi-stage value chain can split across several locations.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A multi-stage value chain can split across several locations.

02

STAGE 02

WEBER'S LEAST-COST THEORY IN SIX ANALYTICAL STEPS

WEBER'S LEAST-COST THEORY IN SIX ANALYTICAL STEPS1. LOCATE RAW-MATERIAL SOURCES AND MARKET2. CLASSIFY INPUTSUBIQUITOUS OR LOCALISED; PURE OR WEIGHT-LOSING3. COMPARE TRANSPORT BURDEN THROUGH THE MATERIAL INDEX4. FIND TRANSPORT-MINIMUM POINT WITHIN THE LOCATIONAL TRIANGLE5. TEST LABOUR SAVING AGAINST ADDED TRANSPORT COST6. TEST AGGLOMERATION SAVING AGAINST CONGESTION AND DISPERSION COST

1. LOCATE raw-material sources and market

2. CLASSIFY inputs: ubiquitous or localised; pure or weight-losing

3. COMPARE transport burden through the material index

4. FIND transport-minimum point within the locational triangle

DEFINITION / COMPONENTS

- 1. LOCATE raw-material sources and market
- 2. CLASSIFY inputs: ubiquitous or localised; pure or weight-losing
- 3. COMPARE transport burden through the material index
- 4. FIND transport-minimum point within the locational triangle

TRIGGER / MECHANISM

- 5. TEST labour saving against added transport cost
- 6. TEST agglomeration saving against congestion and dispersion cost
- LEAST TOTAL COST LOCATION
- TRANSPORT COST + LABOUR COST + AGGLOMERATION ECONOMY

EFFECT / INTERVENTION

- MODERN QUALIFICATION containers + roads + reliable power + data + regulation + value chains alter cost surfaces,
- but distance, weight, time and reliability still matter.
- Weber explains a baseline mechanism, not every final location.

MECHANISM / VERDICT — Weber explains a baseline mechanism, not every final location.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MODERN QUALIFICATION containers + roads + reliable power + data + regulation + value chains alter cost surfaces, but distance, weight, time and reliability still matter.

03

STAGE 03

MAJOR WORLD INDUSTRIAL REGIONS AND THEIR GEOGRAPHIC MECHANISMS

MAJOR WORLD INDUSTRIAL REGIONS AND THEIR GEOGRAPHIC MECHANISMSNE UNITED STATESMARKET + GREAT LAKES + TRANSPORT + SKILLSNW EUROPECOAL LEGACY + RHINE LINKS + DENSE MARKETSN ITALYOLD INDUSTRY + SMES + EUROPEAN NETWORKSJAPAN PACIFIC BELT

REGION	MECHANISM
NE United States	market + Great Lakes + transport + skills
Ruhr / NW Europe	coal legacy + Rhine links + dense markets
Midlands / N Italy	old industry + SMEs + European networks
Japan Pacific Belt	ports + imported inputs + coastal markets
China coastal belt	ports + labour + FDI + supplier depth

SYSTEM / DEFINITION

- NE United States
- market + Great Lakes + transport + skills
- Ruhr / NW Europe
- coal legacy + Rhine links + dense markets
- Midlands / N Italy

PROCESS / MECHANISM

- old industry + SMEs + European networks
- Japan Pacific Belt
- ports + imported inputs + coastal markets
- China coastal belt
- ports + labour + FDI + supplier depth

SPATIAL / INDIA PATTERN

- South Korea corridor
- ports + chaebol networks + technology
- COMMON EVOLUTION resource or port advantage → firm concentration → suppliers and labour
- infrastructure and knowledge → diversification → industrial inertia
- RESTRUCTURING PRESSURE old plant + pollution + land cost + global competition

HAZARD / LIMIT / RESPONSE

- upgrading, decline, relocation or service transition
- The same label hides different factor combinations and historical paths.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: The same label hides different factor combinations and historical paths.

04

STAGE 04

INDIA'S LAYERED INDUSTRIAL MAP: BELTS, CLUSTERS AND CORRIDORS

INDIA'S LAYERED INDUSTRIAL MAPBELTS, CLUSTERS AND CORRIDORSINDIA'S LAYERED INDUSTRIAL GEOGRAPHYCHOTA NAGPURCOAL + ORE + RAILSTEEL, POWER, ENGINEERINGPORT + MARKET + CAPITALDIVERSIFIED INDUSTRY

INDIA'S LAYERED INDUSTRIAL GEOGRAPHY

Chota Nagpur / Damodar : coal + ore + rail → steel, power, engineering

Mumbai-Pune : port + market + capital → diversified industry

Ahmedabad-Vadodara : cotton + enterprise → textile, chemical, engineering

SYSTEM / DEFINITION

- INDIA'S LAYERED INDUSTRIAL GEOGRAPHY
- Chota Nagpur / Damodar : coal + ore + rail → steel, power, engineering

PROCESS / MECHANISM

- Chennai-Bengaluru : ports + skill → auto, electronics, machinery
- Coimbatore-Tiruppur : textile base + SMEs → knitwear, pumps, machinery

SPATIAL / INDIA PATTERN

- CORRIDOR OVERLAY freight spine / expressway / port → planned node → trunk utilities

HAZARD / LIMIT / RESPONSE

- New corridors overlay inherited belts; they do not erase brownfield depth.

- market + Great Lakes + transport + skills
- Ruhr / NW Europe
- coal legacy + Rhine links + dense markets
- Midlands / N Italy

- Japan Pacific Belt
- ports + imported inputs + coastal markets
- China coastal belt
- ports + labour + FDI + supplier depth

- ports + chaebol networks + technology
- COMMON EVOLUTION resource or port advantage → firm concentration → suppliers and labour
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STAGE 04

INDIA'S LAYERED INDUSTRIAL MAP: BELTS, CLUSTERS AND CORRIDORS

INDIA'S LAYERED INDUSTRIAL MAP

BELTS, CLUSTERS AND CORRIDORS

INDIA'S LAYERED INDUSTRIAL GEOGRAPHY

CHOTA NAGPUR

COAL + ORE + RAIL

STEEL, POWER, ENGINEERING

PORT + MARKET + CAPITAL

DIVERSIFIED INDUSTRY

INDIA'S LAYERED INDUSTRIAL GEOGRAPHY

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SYSTEM / DEFINITION

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- Chota Nagpur / Damodar : coal + ore + rail → steel, power, engineering
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PROCESS / MECHANISM

- Chennai-Bengaluru : ports + skill → auto, electronics, machinery
- Coimbatore-Tiruppur : textile base + SMEs → knitwear, pumps, machinery
- Delhi-NCR : market + policy + logistics → diversified cluster
- Visakhapatnam/east : port + mineral hinterland → metal and petro industry

SPATIAL / INDIA PATTERN

- CORRIDOR OVERLAY freight spine / expressway / port → planned node → trunk utilities
- firms and suppliers → labour settlements → possible industrial region
- DMIC west
- AKIC east

HAZARD / LIMIT / RESPONSE

- New corridors overlay inherited belts; they do not erase brownfield depth.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: New corridors overlay inherited belts; they do not erase brownfield depth.

05

STAGE 05

AGGLOMERATION, DEGLOMERATION, INERTIA AND REGIONAL DIVERGENCE

AGGLOMERATION, DEGLOMERATION, INERTIA AND REGIONAL DIVERGENCE

AGGLOMERATION LOOP FIRMS CLUSTER

SHARED SUPPLIERS + LABOUR + SERVICES + KNOWLEDGE

LOWER TRANSACTION COST

MORE FIRMS AND SKILL

DEEPER CLUSTER

CONGESTION THRESHOLD LAND COST + TRAFFIC + POLLUTION +

DEGLOMERATION SELECTED PLANTS OR STAGES MOVE OUTWARD TO CHEAPER

SYSTEM / DEFINITION

AGGLOMERATION LOOP firms cluster → shared suppliers + labour + services + knowledge

lower transaction cost → more firms and skill → deeper cluster

CONGESTION THRESHOLD land cost + traffic + pollution + water stress + labour pressure

PROCESS / MECHANISM

DEGLOMERATION selected plants or stages move outward to cheaper connected sites

INDUSTRIAL INERTIA sunk capital + supplier ties + skill + reputation keep old region active

REGIONAL OUTCOME TEST

SPATIAL / INDIA PATTERN

SPREAD: local suppliers + training + demand + infrastructure diffusion

BACKWASH: talent/capital drain + enclave links + land displacement

A large plant creates convergence only when local capability and linkages grow.

HAZARD / LIMIT / RESPONSE

DEGLOMERATION selected plants or stages move outward to cheaper connected sites

SYSTEM / DEFINITION

- AGGLOMERATION LOOP firms cluster → shared suppliers + labour + services + knowledge
- lower transaction cost → more firms and skill → deeper cluster
- CONGESTION THRESHOLD land cost + traffic + pollution + water stress + labour pressure

PROCESS / MECHANISM

- DEGLOMERATION selected plants or stages move outward to cheaper connected sites
- INDUSTRIAL INERTIA sunk capital + supplier ties + skill + reputation keep old region active
- REGIONAL OUTCOME TEST

SPATIAL / INDIA PATTERN

- SPREAD: local suppliers + training + demand + infrastructure diffusion
- BACKWASH: talent/capital drain + enclave links + land displacement
- A large plant creates convergence only when local capability and linkages grow.

HAZARD / LIMIT / RESPONSE

- DEGLOMERATION selected plants or stages move outward to cheaper connected sites

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A large plant creates convergence only when local capability and linkages grow.

06

STAGE 06

STATE, CORRIDORS, PLI AND THE INDUSTRIAL INTERVENTION STACK

STATE, CORRIDORS, PLI AND THE INDUSTRIAL INTERVENTION STACK

NATIONAL POLICY

MINISTRIES DEFINE FRAMEWORKS AND INCENTIVES

CORRIDOR GOVERNANCE

SPVS PLAN NODES AND TRUNK SYSTEMS

STATE AGENCIES

LAND + UTILITIES + APPROVALS + LOCAL INDUSTRIAL ESTATES

NETWORK ACTORS

NATIONAL POLICY: DPIIT / ministries define frameworks and incentives

CORRIDOR GOVERNANCE: NICDIT / SPVs plan nodes and trunk systems

STATE AGENCIES: land + utilities + approvals + local industrial estates

NETWORK ACTORS: rail freight + ports + roads + power + digital systems

DEFINITION / COMPONENTS

- NATIONAL POLICY: DPIIT / ministries define frameworks and incentives
- CORRIDOR GOVERNANCE: NICDIT / SPVs plan nodes and trunk systems

TRIGGER / MECHANISM

- FIRM ECOSYSTEM: anchor producer + suppliers + skills + testing + finance
- URBAN SUPPORT: housing + mobility + water + waste + social services

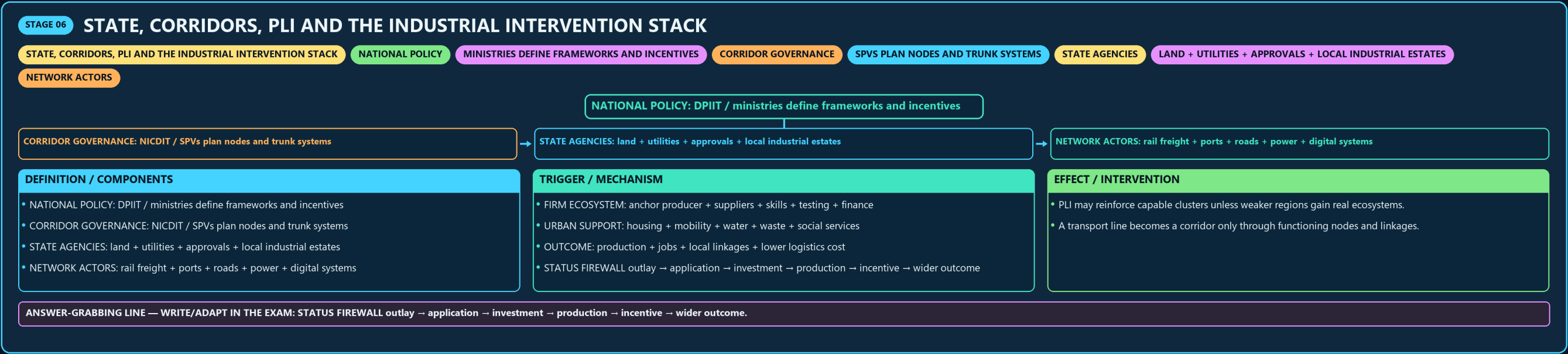
EFFECT / INTERVENTION

- PLI may reinforce capable clusters unless weaker regions gain real ecosystems.
- A transport line becomes a corridor only through functioning nodes and linkages.

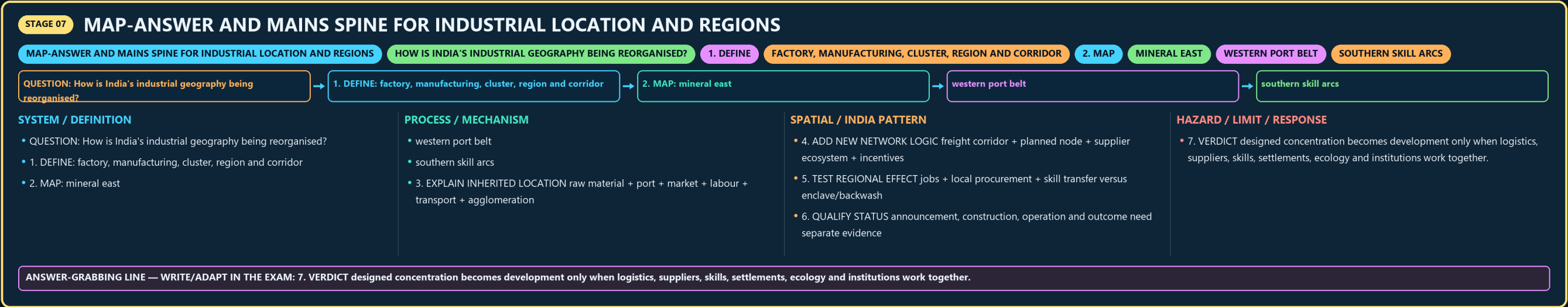
AGGLOMERATION ECONOMY cluster + shared suppliers + labour + services + knowledge • lower transaction cost → more firms and skill → deeper cluster • CONGESTION THRESHOLD land cost + traffic + pollution + water stress + labour pressure	DECONCENTRATION selected plants or stages move outward to cheaper, connected sites • INDUSTRIAL INERTIA sunk capital + supplier ties + skill + reputation keep old region active • REGIONAL OUTCOME TEST	DECONCENTRATION selected plants or stages move outward to cheaper, connected sites • BACKWASH: talent/capital drain + enclave links + land displacement • A large plant creates convergence only when local capability and linkages grow.	DECONCENTRATION selected plants or stages move outward to cheaper, connected sites
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ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: A large plant creates convergence only when local capability and linkages grow.

06



07



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